Ruiming Guo

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Research Interest

- Computational Sensing and Imaging, Sampling Theory, Sensing-Inference Co-Design.
- Machine-Centric Acquisition, Listening and Perception.

Overview: My research aims to advance acquisition-recovery co-design to redefine the limits of computational sensing. At its core is a key insight: structured quantization noise enables *exact* smooth signal recovery where traditional quantized representations fail. Harnessing this principle reshapes long-standing paradigms across digital acquisition, representation, and processing.

Career History

Research Associate (08/2022 – present)
Department of Electrical and Electronic Engineering
Imperial College London
Mentor: Ayush Bhandari

Research Fellow (08/2021 – 08/2022)
Department of Electronic Engineering
The Chinese University of Hong Kong
Mentor: Thierry Blu

Education

The Chinese University of Hong Kong, Hong Kong
Ph.D. in Electronic Engineering, 2021
GPA: 3.93/4.00
Advisor: Thierry Blu

Sichuan University, Chengdu, China
B.S. in Electronic and Information Engineering, 2017
Overall ranking: 1/139
Advisor: Xiaohai He

Awards and Honors

Best Paper Award
International Conference on Sampling Theory and Applications (SampTA), 2025

Postgraduate Student Research Award
Department of Electronic Engineering, The Chinese University of Hong Kong, 2022

Tutor Commendation
The Chinese University of Hong Kong, 2020 – 2021

Hong Kong Postgraduate Scholarship
The Chinese University of Hong Kong, 2017 – 2021

National Scholarship (comprehensive ranking 2/251, top 1%)
Ministry of Education of China, 2014

First Prize, Sichuan Division
The 6th Chinese Mathematics Competitions (CMC), 2014

First Prize (top 2%)
Mathematics Competition of Sichuan University, 2014

Outstanding Student Award (top 5%)
Sichuan University, 2014 and 2015

Invited Talks and Tutorials

Keynote: *Towards Machine-Centric Time-Resolved Imaging: Beyond the Limits of Linear Perception*
ICIP 2026 Workshop on Time-Resolved Computational Imaging (09/2026)

Tutorial: *Unlimited Sensing: From Theoretical Foundations to Practical Impact*
IEEE International Conference on Acoustics, Speech and Signal Processing (ICASSP), 04/2025

Tutorial: *Computational Sensing: From Shannon Sampling to Hardware–Software Co-Design*
European Signal Processing Conference (EUSIPCO), 08/2024

Talk: *Sub-Nyquist USF Spectral Estimation: K Frequencies with $6K + 4$ Modulo Samples*
Imperial College London, 02/2024

Talk: *Unlimited Sensing Framework: Digitization Beyond the Shannon–Nyquist Philosophy*
The Chinese University of Hong Kong, 10/2023; Zhejiang University, 07/2023

Talk: *Geometry Inference from Modulo Signal Space*
International Symposium on Computational Sensing, Luxembourg, 06/2023

Technology Demonstrations

GERF 2026: *Digital Revolution in 2026*
The Great Exhibition Road Festival, 06/2026.

ICASSP 2026: *Unlimited Sensing Radar: Enhancing Resolution via Modulo ADCs*
IEEE Intl. Conf. on Acoustics, Speech and Signal Processing (ICASSP) Show & Tell Demonstration, 05/2026.

ICASSP 2026: *Event-Driven Radar: One-Bit Spectral Super-Resolution and Hardware*
IEEE Intl. Conf. on Acoustics, Speech and Signal Processing (ICASSP) Show & Tell Demonstration, 05/2026.

SampTA 2025: *Unlimited Sensing: From Concept to Impact*
Intl. Conf. on Sampling Theory and Applications (SampTA) Show & Tell Demonstration, 07/2025.

ICASSP 2025: *Hardware Demo of Sub-Nyquist Unlimited Sampling of Multiband Signals*
IEEE Intl. Conf. on Acoustics, Speech and Signal Processing (ICASSP) Show & Tell Demonstration, 04/2025.

ICASSP 2025: *Spectral Estimation with Modulo ADCs: Enhancing Resolution with Sparsity*
IEEE Intl. Conf. on Acoustics, Speech and Signal Processing (ICASSP) Show & Tell Demonstration, 04/2025.

ICASSP 2024: *Sub-Nyquist Frequency Estimation via Multi-Channel Unlimited Sampling ADCs*
IEEE Intl. Conf. on Acoustics, Speech and Signal Processing (ICASSP) Show & Tell

Demonstration, 04/2024.

Teaching

ELEC 70085: *Computational Sensing and Imaging* (Imperial College London)
Spring 2023 – 2026

ELEC 70077: *Digital Signal Processing* (Imperial College London)
Fall 2023 – 2026

ENGG 2420A: *Complex Analysis and Differential Equations* (CUHK)
Fall 2017 – 2020

ENGG 2740A: *Differential Equations* (CUHK)
Spring 2021

ELEG 4502: *Digital Image Processing* (CUHK)
Spring 2019

ENGG 4502: *Digital Signal Processing* (CUHK)
Spring 2017 – 2018

Publications

* denotes co-first authors.

Journal Papers

[J10] R. Guo, Y. Zhu, and A. Bhandari, "Re-thinking digital audio via unlimited sensing: Theory, algorithms, and hardware validation," *IEEE Trans Audio Speech Lang Process*, pp. 1–16, 2026

[J9] R. Guo and A. Bhandari, "Unlocking off-the-grid sparse recovery with unlimited sensing: Simultaneous super-resolution in time and amplitude," *IEEE J. Sel. Topics Signal Process.*, pp. 1–13, 2025

[J8] R. Guo and A. Bhandari, "Blind time-of-flight imaging: Sparse deconvolution on the continuum with unknown kernels," *SIAM J. Imaging Sci.*, vol. 18, pp. 1439–1467, May 2025

[J7] R. Guo, Y. Zhu, and A. Bhandari, "Sub-Nyquist USF spectral estimation: K frequencies with $6K + 4$ modulo samples," *IEEE Trans. Sig. Proc.*, vol. 72, pp. 5065–5076, 2024

[J6] R. Guo and T. Blu, "Super-resolving a frequency band," *IEEE Signal Process. Mag.*, vol. 40, pp. 73–77, Nov. 2023

[J5] Y. Li, R. Guo, T. Blu, and H. Zhao, "Robust sparse reconstruction of attenuated acoustic field with unknown range of source," *The Journal of the Acoustical Society of America*, vol. 152, pp. 3523–3534, Dec. 2022

[J4] R. Guo, Y. Li, T. Blu, and H. Zhao, "Vector-FRI recovery of multi-sensor measurements," *IEEE Trans. Sig. Proc.*, vol. 70, pp. 4369–4380, Sept. 2022

[J3] Y. Li, R. Guo, T. Blu, and H. Zhao, "Generic FRI-based DOA estimation: A model-fitting method," *IEEE Trans. Sig. Proc.*, vol. 69, pp. 4102–4115, June 2021

[J2] R. Guo and T. Blu, "Exploring the geometry of one-dimensional signals," *IEEE Trans. Sig. Proc.*, vol. 69, pp. 5299–5312, 2021

[J1] R. Guo and T. Blu, "FRI sensing: Retrieving the trajectory of a mobile sensor from its temporal samples," *IEEE Trans. Sig. Proc.*, vol. 68, pp. 5533–5545, 2020

Conference Papers

[C20] V. Pavlíček, R. Guo, and A. Bhandari, "Unlimited sensing of sinusoidal plane waves," in *European Sig. Proc. Conf. (EUSIPCO)*, 2026

[C19] R. Guo and A. Bhandari, "Asynchronous multi-channel usf: Modified crt for modulo unfolding," in *European Sig. Proc. Conf. (EUSIPCO)*, 2026

[C18] R. Guo and A. Bhandari, "Digital audio via unlimited sensing: Overflow overcomes clipping and overflow," in *European Sig. Proc. Conf. (EUSIPCO)*, 2026

[C17] V. Pavlíček, R. Guo, and A. Bhandari, "Towards event-driven radars: Spectral super-resolution and hardware," in *IEEE Intl. Conf. on Acoustics, Speech and Signal Processing (ICASSP)*, pp. 776–780, IEEE, May 2026

[C16] R. Guo, V. Pavlíček, and A. Bhandari, "Enhancing Doppler and FMCW radars via unlimited sensing," in *IEEE Intl. Conf. on Acoustics, Speech and Signal Processing (ICASSP)*, pp. 761–765, IEEE, May 2026

[C15] R. Guo and A. Bhandari, "HoD-FP algorithm for unlimited sensing: Where time meets frequency," in *IEEE International Workshop on Computational Advances in Multi-Sensor Adaptive Processing (CAMSAP)*, pp. 116–120, IEEE, Dec. 2025

[C14] Y. Zhu, R. Guo, and A. Bhandari, "Ironing the modulo folds: Unlimited sensing with 100x bandwidth expansion," in *Intl. Conf. on Sampling Theory and Applications (SampTA)*, pp. 1–5, IEEE, July 2025. **Best Paper Award**

[C13] R. Guo and A. Bhandari, "USF spectral estimation: Prevalence of Gaussian Cramér-Rao bounds despite modulo folding," in *IEEE Statistical Signal Processing Workshop (SSP)*, pp. 51–55, IEEE, June 2025

[C12] R. Guo and A. Bhandari, "Spectrum blind unlimited sampling of multi-band signals," in *IEEE Intl. Conf. on Acoustics, Speech and Signal Processing (ICASSP)*, pp. 1–5, IEEE, Apr. 2025

[C11] R. Guo, G. Shtendel, and A. Bhandari, "Multiband unlimited sampling: Super-Nyquist or sub-Nyquist, that is the question!," in *IEEE Intl. Conf. on Acoustics, Speech and Signal Processing (ICASSP)*, pp. 1–5, IEEE, Apr. 2025

[C10] R. Guo, Y. Zhu, and A. Bhandari, "Event-driven Prony: Towards asynchronous spectral estimation," in *IEEE Intl. Conf. on Acoustics, Speech and Signal Processing (ICASSP)*, pp. 1–5, IEEE, Apr. 2025

[C9] Q. Dai, R. Guo, and T. Blu, "Fast and robust high resolution frequency estimation of damped signals," in *IEEE Intl. Conf. on Acoustics, Speech and Signal Processing (ICASSP)*, pp. 1–5, IEEE, Apr. 2025

[C8] R. Guo and A. Bhandari, "ED- FreEst: Event - driven frequency estimation," in *European Sig. Proc. Conf. (EUSIPCO)*, pp. 867–871, IEEE, Aug. 2024

[C7] V. Pavlíček, R. Guo, and A. Bhandari, "Bits, channels, frequencies and unlimited sensing: Pushing the limits of Sub-Nyquist Prony," in *European Sig. Proc. Conf. (EUSIPCO)*, pp. 2462–2466, IEEE, Aug. 2024

[C6] Q. Dai, R. Guo, and T. Blu, "Super-resolving single radar target with an exact and simple formula," in *Intl. Tech. Conf. on Circuits/Systems, Computers, and Communications (ITC-CSCC)*, pp. 1–5, IEEE, July 2024

[C5] Y. Zhu, R. Guo, P. Zhang, and A. Bhandari, "Frequency estimation via sub-Nyquist unlimited sampling," in *IEEE Intl. Conf. on Acoustics, Speech and Signal Processing (ICASSP)*, IEEE, Apr. 2024

[C4] R. Guo and A. Bhandari, "ITER-SIS: Robust unlimited sampling via iterative signal sieving," in *IEEE Intl. Conf. on Acoustics, Speech and Signal Processing (ICASSP)*, June 2023

[C3] R. Guo and A. Bhandari, "Unlimited sampling of FRI signals independent of sampling rate," in *IEEE Intl. Conf. on Acoustics, Speech and Signal Processing (ICASSP)*, June 2023

[C2] W. Wang, R. Guo, Y. Tian, and W. Yang, "CFSNet: Toward a controllable feature space for image restoration," in *IEEE Intl. Conf. on Computer Vision (ICCV)*, pp. 4139–4148, IEEE, Oct. 2019

[C1] R. Guo and T. Blu, "FRI sensing: Sampling images along unknown curves," in *IEEE Intl. Conf. on Acoustics, Speech and Signal Processing (ICASSP)*, pp. 5132–5136, IEEE, May 2019